THE FUNCTORS CAN AND H^0 ARE QUASI-INVERSE EQUIVALENCES FOR THE GLUED CATEGORY OF A FLAT COVER WITH A FINITELY GENERATED IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled proposition environment at source lines 24812-24821 (canonical tag 05ER) and the counted proof is the proof environment at source lines 24823-24879, which immediately follows it and is its own proof.

Editorial note. The proof displays the comparison morphism and, later, a commutative diagram with exact rows in the author's own Xy-pic markup, both of which are compiled from that markup with the shared preamble.

Editorial note. The proof ends with the parenthetical remark that a posteriori the module Q vanishes; the remark is the author's own and is preserved verbatim inside the counted span.

Editorial note. The chapter file pinned-original-source.tex carries only an input of the shared preamble on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus the native theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

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Allowed standard prerequisites (editorial).

- The composite of the functor Can with H^0 is isomorphic to the identity functor on the category of modules. used at proof lines 24823-24826. Disposes of one of the two composites, so that only the composite starting from a glued object has to be analysed.
- After inverting one of the generators the comparison morphism becomes an isomorphism, and its kernel and cokernel are then systems concentrated in the first component. used at proof lines 24841-24846. Reduces the analysis of the kernel and cokernel of the comparison morphism to systems of the special form with only the S-component present.
- A system of the form with only the first component present has the property that H^0 of it is that component, and H^0 is left exact, so the exactness computation gives that the kernel vanishes. used at proof lines 24846-24849. Shows that the comparison morphism is injective, which is the first half of the proof that it is an isomorphism.

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- For an I-power torsion module the canonical map from the tensor product with S is an isomorphism, and extensions of the module by the image of H^0 can be realized inside the glued category. used at proof lines 24856-24860. Produces the commutative diagram with exact rows whose middle term gives a module E with Can(E) isomorphic to the glued object.

Source reference. Original lemma 05EM: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24729–24729. The reference records that one of the two composites is the identity functor, used at proof lines 24824-24826; the analysis of the other composite, which is the content of the proof, is inside the counted span.

Source reference. Original lemma 05EQ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24792–24792. The reference is combined with the next one at proof lines 24842-24844 to show that the comparison morphism becomes an isomorphism after inverting a generator; the conclusions drawn from that localization are inside the counted span.

Source reference. Original lemma 05EP: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24768–24768. The reference is the second of the two lemmas applied at proof lines 24842-24844 to localize the comparison morphism; the ensuing injectivity and exactness arguments are inside the counted span.

Source reference. Original lemma 05EG: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24477–24477. The reference supplies the existence of the commutative diagram with exact rows used at proof lines 24858-24874 to produce the module E; the argument that this E maps under Can to the given glued object is inside the counted span.

Proposition 0.2. *Assume $\varphi : R \to S$ is a flat ring map and $I = (f_1, \ldots, f_t) \subset R$ is an ideal such that $R/I \to S/IS$ is an isomorphism. Then Can and H^0 are quasi-inverse equivalences of categories*

$$Mod_R = Glue(R \to S, f_1, \ldots, f_t)$$

Proof. We have already seen that $H^0 \circ \text{Can}$ is isomorphic to the identity functor, see Lemma 05EM. Consider an object $\mathbf{M} = (M', M_i, \alpha_i, \alpha_{ij})$ of $\text{Glue}(R \to S, f_1, \ldots, f_t)$. We get a natural morphism

$$\Psi : (H^0(\mathbf{M}) \otimes_R S, H^0(\mathbf{M})_{f_i}, \text{can}_i, \text{can}_{ij}) \longrightarrow (M', M_i, \alpha_i, \alpha_{ij}).$$

Namely, by definition $H^0(\mathbf{M})$ comes equipped with compatible R-module maps $H^0(\mathbf{M}) \to M'$ and $H^0(\mathbf{M}) \to M_i$. We have to show that this map is an isomorphism.

Pick an index i and set $R' = R_{f_i}$. Combining Lemmas 05EQ and 05EP we see that $\Psi \otimes_R R'$ is an isomorphism. Hence the kernel, resp. cokernel of Ψ is a system of the form $(K, 0, 0, 0)$, resp. $(Q, 0, 0, 0)$. Note that $H^0((K, 0, 0, 0)) = K$, that H^0 is left exact, and that by construction $H^0(\Psi)$ is bijective. Hence we see $K = 0$, i.e., the kernel of Ψ is zero.

The conclusion of the above is that we obtain a short exact sequence

$$0 \to H^0(\mathbf{M}) \otimes_R S \to M' \to Q \to 0$$

and that $M_i = H^0(\mathbf{M})_{f_i}$. Note that we may think of Q as an R-module which is I-power torsion so that $Q = Q \otimes_R S$. By Lemma 05EG we see that there exists a

commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & H^0(\mathbf{M}) & \longrightarrow & E & \longrightarrow & Q & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & H^0(\mathbf{M}) \otimes_R S & \longrightarrow & M' & \longrightarrow & Q & \longrightarrow & 0 \end{array}$$

with exact rows. This clearly determines an isomorphism $\mathrm{Can}(E) \to (M', M_i, \alpha_i, \alpha_{ij})$ in the category $\mathrm{Glue}(R \to S, f_1, \ldots, f_t)$ and we win. (Of course, a posteriori we have $Q = 0$.) $\square$